

Indian Institute of Technology Jodhpur

Department of Mechanical Engineering, Nagour Road, Karwar, Jodhpur 342030, Rajasthan, INDIA

Course: Engineering Mechanics

Due Date: 06/01/2026

Code: MEL1010

Tutorial: Prep

1. You may make acceleration and other measurements using the Physics Toolbox Sensor Suite: <https://youtu.be/tu6sHeJ4Ys> . You may walk/Jog on the inner circle of the jogging/athletic track on IIT Jodhpur at a uniform speed comfortable to you as far possible and make measurements for three rounds. Download the CSV data and compute the number of steps and total distance using your own approach.
2. You may also use any of the PEDOMETER apps available and compute the number of steps and total distance covered. Compare the result with that obtained above using Physics Toolbox Sensor Suite. Comment on the accuracy of the results.
3. Measure the acceleration due to gravity using the sensor in your mobile phone. Give the details and specifications and sensitivity, accuracy of the accelerometer in your mobile phone. Note the latitude and longitude of Jodhpur. Use any of the formulas you feel is appropriate from the website (https://en.wikipedia.org/wiki/Theoretical_gravity) to compute “g” for Jodhpur campus of IIT Jodhpur. How accurate is your measurement with respect to the computed value.